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**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2023-2024)

Test Pattern

NEET (UG)

MINOR

21-05-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MEA & MEPS-I (A)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill 0012345678.

Test Booklet Code

This Booklet contains 28 pages.

E4

Do not open this Test Booklet until you are asked to do so.

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- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. 50 questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

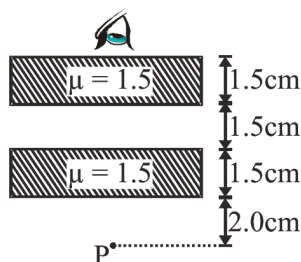
Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in Pre-Medical 2024

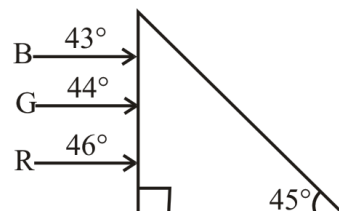
SECTION-A (PHYSICS)

- Unpolarised light of intensity 32 Wm^{-2} passes through three polarizer such that transmission axes of the first and second polarizer makes an angle 30° with each other and the transmission axis of the last polarizer is perpendicular to that of the first. The intensity of final emerging light will be :-
 - 32 Wm^{-2}
 - 3 Wm^{-2}
 - 8 Wm^{-2}
 - 4 Wm^{-2}
- If light is polarised by reflection, then the angle between reflected and refracted light beam is :-
 - π
 - $\frac{\pi}{2}$
 - 2π
 - $\frac{\pi}{4}$
- When the angle of incidence on a material is 60° , the reflected light is completely polarized. The velocity of refracted ray inside the material is (in ms^{-1})
 - 3×10^8
 - $\left(\frac{3}{\sqrt{2}}\right) \times 10^8$
 - $\sqrt{3} \times 10^8$
 - 0.5×10^8
- Where should an object of height 5cm be placed In front of a concave mirror of focal length 10cm so that a real Image of height 1cm is obtained :-
 - 48 cm
 - 60 cm
 - 15 cm
 - 30 cm

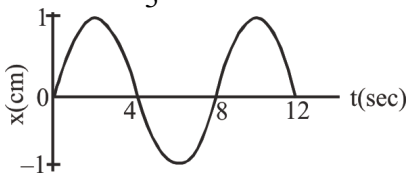
- The image of point P when viewed from top of the slabs will be :-



- 2.0 cm above P
 - 1.5 cm above P
 - 2.0 cm below P
 - 1 cm above P
- A point source of light is placed 4 m below the surface of water of $\mu = \frac{5}{3}$. The minimum diameter of a disc, which should be placed over the source, on the surface of water to cut off all light coming out of water, is
 - 1 m
 - 6 m
 - 4 m
 - 3 m
 - A ray of light is incident normally on one of the faces of a prism of apex angle 30° and refractive index $\sqrt{2}$. The angle of deviation of the ray is:-
 - 0°
 - 12.5°
 - 15°
 - 22.5°
 - Figure shows a mixture of blue, green and red coloured rays incident normally on a right angled prism. The critical angles of the material of the prism for red, green and blue are 46° , 44° and 43° respectively. The arrangement will separate.



- Green colour from red and blue
- All the three colours
- Red colour from blue and green
- Blue colour from red and green

9. If refractive index of red, violet and yellow lights are 1.42, 1.62 and 1.50 respectively for a medium. Its dispersive power will be :-
 (1) 0.4 (2) 0.3 (3) 0.2 (4) 0.1
10. A concave lens of glass, refractive index 1.5, has both surfaces of same radius of curvature R. On immersion in a medium of refractive index 1.75, it will behave as a :-
 (1) Convergent lens of focal length 3.5 R
 (2) Convergent lens of focal length 3.0 R
 (3) Divergent lens of focal length 3.5 R
 (4) Divergent lens of focal length 3.0 R
11. A far sighted person has a near point of 60 cm. What power lens should be used for eye glasses such that the person can read a book at a distance of 25 cm. :-
 (1) -6.66 D (2) +6.66 D
 (3) +2.33 D (4) -2.33 D
12. A compound microscope has an eye-piece of focal length 10 cm and objective of focal length 4 cm. Calculate the magnifying power if final image of an object kept at a distance of 5cm, forms at least distance of distinct vision :-
 (1) +12 (2) -12 (3) +14 (4) -14
13. **Assertion** : When a particle is at extreme position performing SHM, its momentum is equal to zero.
Reason : At extreme position the velocity of particle performing SHM is equal to zero.
 (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
 (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
 (3) Assertion is true but Reason is false.
 (4) Assertion is false but Reason is true.
14. The x-t graph of a particle undergoing simple harmonic motion is shown below. The acceleration of the particle at $t = \frac{4}{3}$ s is :-

 (1) $\frac{\sqrt{3}}{32}\pi^2 \text{ cm/s}^2$ (2) $-\frac{\pi^2}{32} \text{ cm/s}^2$
 (3) $\frac{\pi^2}{32} \text{ cm/s}^2$ (4) $-\frac{\sqrt{3}}{32}\pi^2 \text{ cm/s}^2$
15. The time period of a particle in simple harmonic motion is 8 seconds. At $t = 0$, it is at the mean position. The ratio of the distances travelled by it in the first and second seconds is :-
 (1) $\frac{1}{2}$ (2) $\frac{1}{\sqrt{2}}$
 (3) $\frac{1}{(\sqrt{2}-1)}$ (4) $\frac{1}{\sqrt{3}}$
16. Starting from the mean position, a body oscillates simple harmonically with a period of 2s. After what time will its kinetic energy be 75% of the total energy for the first time?
 (1) 1/12 sec (2) 1/6 sec
 (3) 1/4 sec (4) 1/3 sec
17. If x-t relation is given by $x = A \sin\left(\omega t + \frac{\pi}{4}\right)$ then find time when speed will be zero first time where T is time period of motion :-
 (1) $\frac{T}{6}$ (2) $\frac{T}{8}$
 (3) $\frac{T}{4}$ (4) $\frac{T}{12}$
18. A particle executing SHM of time period T, time taken by particle to complete $\frac{3}{8}$ oscillation will be :
 (1) $\frac{5T}{12}$ (2) $\frac{3T}{8}$
 (3) $\frac{5T}{8}$ (4) $\frac{T}{12}$

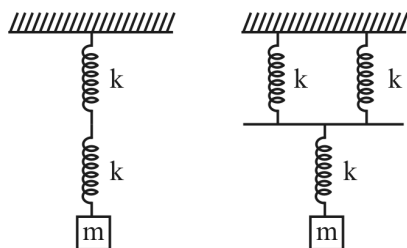
19. A simple pendulum performs S.H.M about $x = 0$ with an amplitude A and time-period T . The speed of pendulum at $x = \frac{A}{2}$ will be :-

- (1) $\frac{A}{T}$
- (2) $\frac{\sqrt{3}\pi A}{T}$
- (3) $\frac{\sqrt{3}(\pi A)}{2T}$
- (4) $\frac{3\pi^2 A}{T}$

20. The kinetic energy of SHM is $1/n$ time its potential energy. If the amplitude of the SHM be A , then what is the displacement of the particle?

- (1) nA
- (2) $\sqrt{\frac{n+1}{n}} A$
- (3) $\sqrt{\frac{n}{n+1}} A$
- (4) $\frac{A}{n}$

21. Some springs are combined in series and parallel arrangement as shown in the figure and a mass m is suspended from them. The ratio of their frequencies will be :-



- (1) 1 : 1
- (2) 2 : 1
- (3) $\sqrt{3} : 2$
- (4) 4 : 1

22. In damped oscillations, the amplitude of oscillations is reduced to one-third of its initial value a_0 at the end of 100 oscillations. When the oscillator completes 200 oscillations, its amplitude must be

- (1) $a_0/2$
- (2) $a_0/4$
- (3) $a_0/6$
- (4) $a_0/9$

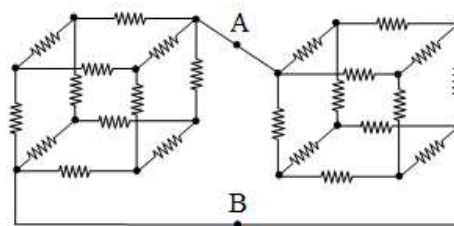
23. At room temperature 27°C , the resistance of heating element is 100Ω then at what temperature its resistance will become 117Ω (temperature coefficient of the element is 1.7×10^{-4} per $^\circ\text{C}$)

- (1) 1027°C
- (2) 1000°C
- (3) 927°C
- (4) 900°C

24. Resistance of a conductor depends on

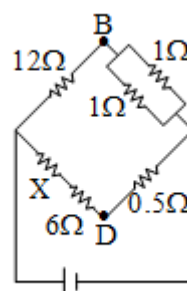
- (1) Size of conductor
- (2) Temperature of conductor
- (3) Geometry of conductor
- (4) All of above

25. Consider the network of equal resistances (each R) shown in **Figure**. Then the effective resistance between points A and B is -



- (1) $(5/3) R$
- (2) $(5/6) R$
- (3) $(5/12) R$
- (4) None of these

26. For what value of unknown resistance X the potential difference between B and D will be zero in arrangement of adjacent figure -



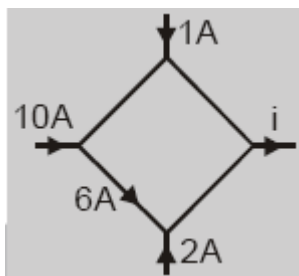
- (1) 4Ω
- (2) 3Ω
- (3) 2Ω
- (4) 6Ω

English

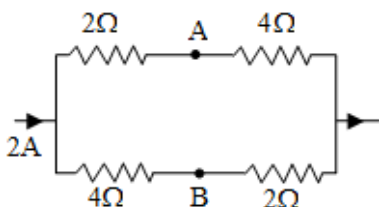
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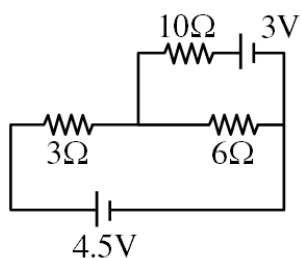
27. Value of current i in the following circuit is :-



- (1) 13 A
(2) 12 A
(3) 9 A
(4) none of the above
28. In adjacent circuit $V_A - V_B$ will be -



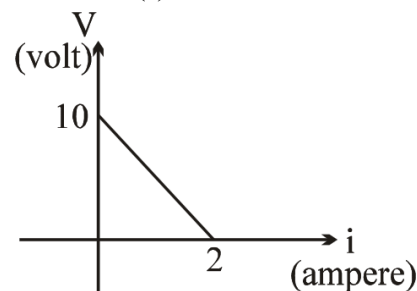
- (1) 4 V (2) 2 V
(3) 3 V (4) zero
29. Find the current through the $10\ \Omega$ resistor shown in figure:



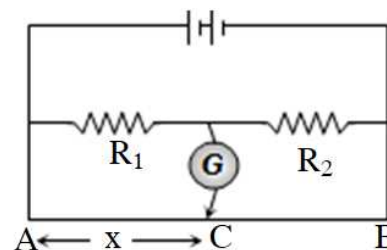
- (1) zero
(2) 1A
(3) 2A
(4) 5A
30. The internal resistance of a cell of emf 12 V is $5 \times 10^{-2}\ \Omega$. It is connected across an unknown resistance. Voltage across the cell, when a current of 60 A is drawn from it, is

- (1) 15 V (2) 12 V (3) 9 V (4) 6 V

31. A battery of emf E and internal resistance r is connected across a resistance R . Resistance R can be adjusted to any value greater than or equal to zero. A graph is plotted between the current (i) passing through the resistance and potential difference (V) across it. Select the correct alternative(s).



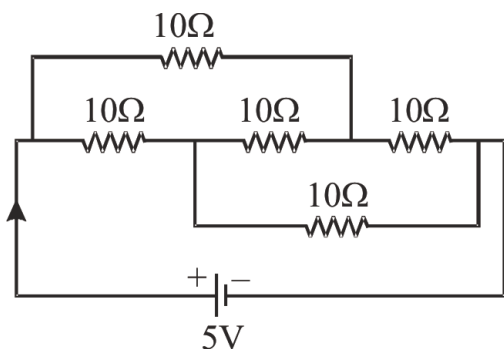
- (1) internal resistance of battery is $5\ \Omega$
(2) emf of the battery is 20V
(3) maximum current which can be taken from the battery is 4A
(4) $V-i$ graph can never be a straight line as shown in figure.
32. In the shown arrangement of the experiment of the meter bridge if AC corresponding to null deflection of galvanometer is x , what would be its value if the radius of the wire AB is doubled



- (1) x
(2) $x/4$
(3) $4x$
(4) $2x$
33. 100 cells, each of emf 5 V and internal resistance $1\ \Omega$ are to be arranged so as to produce maximum current in a $25\ \Omega$ resistance. Each row is to contain equal number of cells. The number of rows should be

- (1) 2 (2) 4 (3) 5 (4) 10

34. The current I drawn from the 5 volt source will be:-



- (1) 0.17 A (2) 0.33 A
(3) 0.5 A (4) 0.67 A
35. Internal resistance depends on
- (1) Concentration of electrolyte
(2) Temperature
(3) Separation between electrodes
(4) All of the above

SECTION-B (PHYSICS)

36. **STATEMENT-1** : The formula connecting u , v and f for a spherical mirror is valid only for mirrors whose sizes are very small compared to their radii of curvature i.e for paraxial ray.
STATEMENT-2 : Laws of reflection are strictly valid for plane surfaces, but not for large spherical surfaces.
- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(3) Statement-1 is True, Statement-2 is False
(4) Statement-1 is False, Statement-2 is True
37. A bucket of total height 60 cm is half filled with a liquid of refractive index 1.5 and half with another liquid of refractive index 2. The apparent depth of the bucket for an observer directly above the bucket-
- (1) 45 cm (2) 30 cm
(3) 35 cm (4) 40 cm

38. Light travels in two media A & B with speeds 1.8×10^8 m/s & 2.4×10^8 m/s respectively. Then the critical angle between them is :-
- (1) $\sin^{-1}(2/3)$ (2) $\tan^{-1}(3/4)$
(3) $\tan^{-1}(2/3)$ (4) $\sin^{-1}(3/4)$
39. For the angle of minimum deviation of a prism to be equal to its refracting angle, the prism must be made of a material whose refractive index :-
- (1) between $\sqrt{2}$ & 1 (2) between 2 & $\sqrt{2}$
(3) less than 1 (4) Greater than 2
40. An object of size 3cm is placed 14cm in front of a concave lens of focal length 21cm. The image distance v , magnification m , image height I . (With reference to the above situation, match the items in Column I with terms in Column II and choose the option given below.)

Column I		Column II	
(A)	v (Image distance in cm)	(1)	0.6
(B)	I (Image height in cm)	(2)	8.4
(C)	m (magnification)	(3)	1.8

- (1) A-1, B-2, C-3 (2) A-3, B-2, C-1
(3) A-2, B-1, C-3 (4) A-2, B-3, C-1
41. An astronomical telescope has an eye piece of focal length 5 cm. If magnifying power of telescope is 14 when final image is formed at infinity. Calculate length of the telescope :-
- (1) 75 cm (2) 9 cm
(3) 50 cm (4) 55 cm
42. The motion of a particle varies with time according to the relation :-
 $y = a(\sin \omega t + \cos \omega t)$, then :
- (1) the motion is oscillatory but not SHM
(2) the motion is SHM with amplitude a
(3) the motion is SHM with amplitude $a\sqrt{2}$
(4) the motion is SHM with amplitude $2a$

43. A particle vibrates in SHM along a straight line. Its greatest acceleration is $5\pi^2 \text{ cm s}^{-2}$ and when its distance from the equilibrium position is 4 cm, the velocity of the particle is $3\pi \text{ cm s}^{-1}$. The amplitude and the period of oscillation of the vibrating particle is :-

- (1) 10 cm, 4 sec (2) 5 cm, 2 sec
(3) 5 cm, 4 sec (4) 10 cm, 2 sec

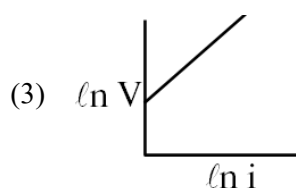
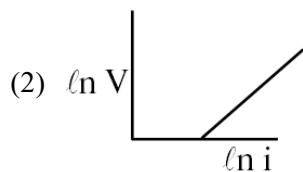
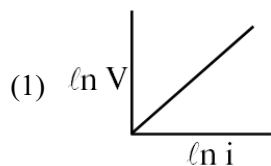
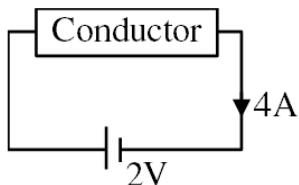
44. Time period (T) and amplitude (A) are same for two particles which undergo SHM along the same line, at one particular instant, one particle is at phase $\frac{3\pi}{2}$ and other is at phase zero. Find the time at which they will cross each other.

- (1) $\frac{4T}{2}$ (2) $\frac{3T}{8}$ (3) $\frac{3T}{4}$ (4) $\frac{3T}{7}$

45. A simple pendulum is executing simple harmonic motion with a time period T. If the length of the pendulum is increased by 21%, the increase in the time period of the pendulum of increased length is:-

- (1) 10% (2) 21% (3) 30% (4) 50%

46. For the given conductor $\ln V - \ln i$ graph is



- (4) None of the above

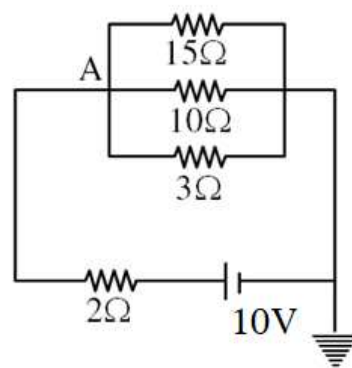
47. Temperature coefficient of resistance for resistances R, 2R and 2R are α , α and 4α respectively, then equivalent temperature coefficient their parallel combination is

- (1) $\frac{7}{4} \alpha$ (2) $\frac{7\alpha}{2}$ (3) 2.5α (4) 2α

48. Which of the following statement is not correct

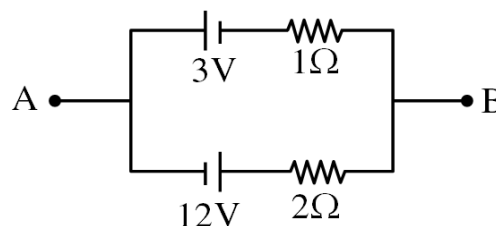
- (1) voltage versus current graph deviates from straight line for a good conductor at high voltage
(2) In consideration of ohm's law, it is assumed that relaxation time is independent of electric field
(3) superconductors are good conductors at room temperature
(4) Resistance of constantan does not change considerably with temperature

49. Potential energy of a charge 2C at the point A is



- (1) 2J (2) 10J
(3) 5J (4) 1J

50. Find potential difference ($V_a - V_b$)



- (1) 6 V (2) -6V
(3) 2V (4) -2V

Topic : COORDINATION COMPOUND, p-BLOCK (UPTO IPE), COMMERCIAL CELLS, SOLID STATE, SURFACE CHEMISTRY [BACK UNIT] : CHEMICAL KINETICS

TG: @Chalnaayaaar

SECTION-A (CHEMISTRY)

51. The stability of +1 oxidation state increases in sequence :-

- (1) $Tl < In < Ga < Al$
- (2) $In < Tl < Ga < Al$
- (3) $Ga < In < Al < Tl$
- (4) $Al < Ga < In < Tl$

52. Ammonia acts as a ligand but ammonium ion does not form complexes because

- (1) NH_3 is a gas while NH_4^+ is in liquid form
- (2) NH_3 undergoes sp^3 hybridisation while NH_4^+ undergoes sp^3d hybridisation
- (3) NH_4^+ ion does not have any lone pair
- (4) NH_4^+ ion has one unpaired electron while NH_3 has two unpaired electrons.

53. Correct match is :

Complexes		No. of unpaired electron	
A	$[CrF_6]^{4-}$	P	5
B	$[MnF_6]^{4-}$	Q	2
C	$[Cr(CN)_6]^{4-}$	R	1
D	$[Mn(CN)_6]^{4-}$	S	4

- (1) (A – S), (B – Q), (C – P), (D – R)
- (2) (A – P), (B – R), (C – Q), (D – S)
- (3) (A – S), (B – P), (C – Q), (D – R)
- (4) (A – R), (B – S), (C – Q), (D – P)

54. The EAN value of the complex $[Fe(CO)_2(NO)_2]$ is:-

- (1) 36
- (2) 34
- (3) 32
- (4) 30

55. Consider the following statements : According to the Werner's theory.

- I. Ligands are connected to the central metal ions by ionic bonds.
- II. Secondary valencies have directional properties
- III. Secondary valencies are non-ionisable

Correct statements are:

- (1) I, II and III are correct
- (2) II and III are correct
- (3) I and III are correct
- (4) I and II are correct

56. The complex $[Co(NH_3)_5(CN)]Cl_2$ and $[Co(NH_3)_5(NC)]Cl_2$ are the examples of :-

- (1) Co-ordination isomerism
- (2) Ionisation isomerism
- (3) Geometrical isomerism
- (4) Linkage isomerism

57. How many geometrical isomers and stereoisomers are possible for $[Pt(Br)(Cl)(I)(NH_2)(NH_3)(Py)]$

- (1) 3 and 15
- (2) 15 and 30
- (3) 4 and 15
- (4) 4 and 30

58. Which of the following is not optically active ?

- (1) $[Co(en)_3]^{3+}$
- (2) $[Cr(ox)_3]^{3-}$
- (3) $cis-[CoCl_2(en)_2]^+$
- (4) $trans-[CoCl_2(en)_2]^+$

59. The incorrect statement about $Ni(CO)_4$ is:

- (1) The bond order of CO in the complex is less than bond order of CO molecule
- (2) The complex is diamagnetic and dsp^2 hybridised
- (3) The bond order of Ni-C bond is greater than one
- (4) The complex cannot act as oxidizing or reducing agent according to sidwick EAN rule.

60. Both geometrical and optical isomerism are shown by :

- (1) $[Co(en)_2Cl_2]^+$
- (2) $[Co(NH_3)_5Cl]^{2+}$
- (3) $[Co(NH_3)_4Cl_2]^+$
- (4) $[Cr(ox)_3]^{3-}$

61. Which of the following order is not correct ?

- (1) $[\text{Cr}(\text{NH}_3)_6]^{3+} > [\text{Ni}(\text{NH}_3)_4\text{Cl}_2] > [\text{V}(\text{CO})_6]$
spin Magnetic moment
- (2) $[\text{Co}(\text{CN})_6]^{3-} > [\text{Co}(\text{NH}_3)_6]^{3+} > [\text{Co}(\text{H}_2\text{O})_6]^{3+}$
CFSE order
- (3) $[\text{Ni}(\text{CO})_4] > [\text{Co}(\text{CO})_4]^- > [\text{Fe}(\text{CO})_4]^{2-}$
M-C π bond strength
- (4) $[\text{Ni}(\text{NH}_3)_4]^{2+} < [\text{Ni}(\text{en})(\text{NH}_3)_2]^{2+} < [\text{Ni}(\text{en})_2]^{2+}$
Thermodynamic stability

62. As per IUPAC nomenclature, the name of the complex $[\text{Co}(\text{H}_2\text{O})_4(\text{NH}_3)_2]\text{Cl}_3$ is:

- (1) tetraaquadiaminacobalt(III) chloride
- (2) tetraaquadiamminecobalt(III) chloride
- (3) diaminetetraaquacobalt(III) chloride
- (4) diamminetetraaquacobalt(III) chloride

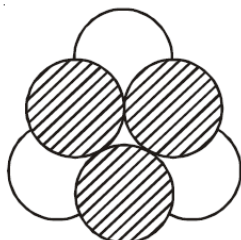
63. Which of the following crystal is represented by $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90^\circ$?

- (1) Orthorhombic (2) Monoclinic
- (3) Triclinic (4) Tetragonal

64. The radius of the Na^+ is 95 pm and that of Cl^- ion is 181 pm. The co-ordination number of Na^+ will be

- (1) 4 (2) 6
- (3) 8 (4) Unpredictable

65. The empty space between the shaded balls and hollow balls as shown in the diagram is called



- (1) Hexagonal void
- (2) Octahedral void
- (3) Tetrahedral void
- (4) Double triangular void

66. Which of the following is incorrect ?

- (1) Tyndall effect provides information about size and shape of disperse particles.
- (2) Mechanical dispersion is used to prepare printer ink.
- (3) Coagulating power of an ion depends on both size and charge of ion.
- (4) Both smog and fog are aerosol.

67. Metal excess defect due to anionic vacancies is observed in:

- (1) AgBr (2) NaCl
- (3) ZnO (4) Metallic solids

68. In solid lattice, the cation has left a site and is located at interstitial position, the lattice defect is called

- (1) Interstitial defect (2) Vacancy defect
- (3) Frenkel defect (4) Schottky defect

69. The number of atoms present in a unit cell of a monoatomic element of a simple cubic lattice, body-centred cubic and face centred cubic respectively are

- (1) 8, 9 and 14 (2) 1, 2 and 4
- (3) 4, 5 and 6 (4) 2, 3 and 5

70. When the substance is placed in magnetic field all the domains get oriented in the direction of magnetic field. Substance is:

- (1) Paramagnetic
- (2) Ferromagnetic
- (3) Ferrimagnetic
- (4) Antiferromagnetic

71. Experimentally it was found that a metal oxide has formula $\text{M}_{0.98}\text{O}_1$. Metal M, is present as M^{2+} and M^{3+} in its oxide. Fraction of the metal which exists as M^{3+} would be :-

- (1) 7.01% (2) 4.08%
- (3) 6.05% (4) 8.12%

72. In the face centred cubic lattice, atom A occupies the corner positions and atom B occupies the face centre positions. If one atom of B is missing from one of the face centred points, the formula of the compound is

- (1) A_2B (2) AB_2 (3) A_2B_2 (4) A_2B_5

73. On adding $AgNO_3$ solution into KI solution, a negatively charged colloidal sol is obtained when they are in :-

- (1) 50 mL of 0.1 M $AgNO_3$ + 50 mL of 0.1 M KI
 (2) 50 mL of 0.1 M $AgNO_3$ + 50 mL of 0.2 M KI
 (3) 50 mL of 0.2 M $AgNO_3$ + 50 mL of 0.1 M KI
 (4) None of these

74. Which of the following acts as an emulsifier?

- (1) Soap (2) Water
 (3) CH_3COOH (4) NaCl

75. How many of the following statements are correct:

- (a) Physical adsorption can be converted into chemical adsorption at high temperature.
 (b) Heterogenous catalysis is an example of chemisorption.
 (c) A catalyst can catalyse a non spontaneous reaction.
 (d) Chemisorption is irreversible.

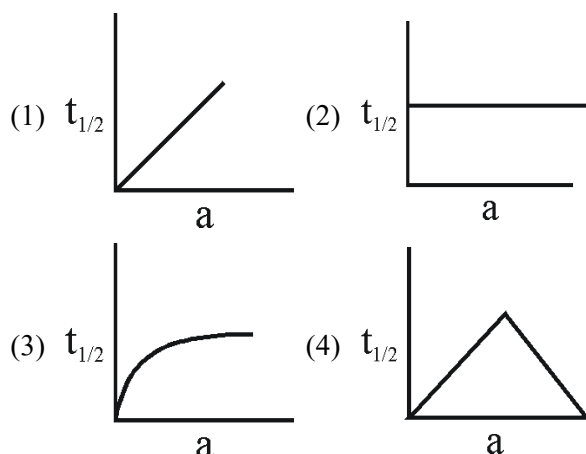
- (1) 1 (2) 2 (3) 3 (4) 4

76. For 1st order reaction find ratio of $t_{99.9\%}$ to that of $t_{50\%}$.

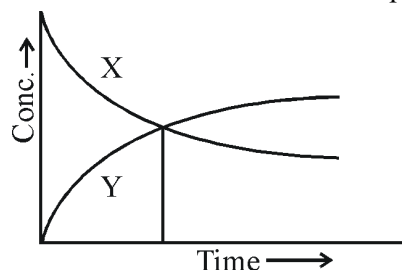
- (1) 10 (2) 3 (3) 4 (4) 17

77. Which of the following curves represents a 1st order reaction-

($t_{1/2}$ = half life), (a = initial concentration)



78. The accompanying figure depicts the change in concentration of species X and Y for the reaction $X \rightarrow Y$, as a function of time. The point of intersection of the two curves represents :



- (1) $t_{1/2}$
 (2) $t_{3/4}$
 (3) $t_{2/3}$
 (4) Data is insufficient to predict

79. An endothermic reaction $A \rightarrow B$ have an activation energy of 15 kcal/mol and the heat of the reaction is 5 kcal/mol. The activation energy of the reaction $B \rightarrow A$ is :

- (1) 20 kcal/mol
 (2) 15 kcal/mol
 (3) 10 kcal/mol
 (4) zero

80. The rate of the elementary reaction, $2NO + O_2 \rightarrow 2NO_2$, when the volume of the reaction vessel is doubled :

- (1) will grow eight times of its initial rate
 (2) reduce to one-eighth of its initial rate
 (3) will grow four times of its initial rate
 (4) reduce to one-fourth of its initial rate

81. The concentration of a reactant in a solution falls from 0.2 M to 0.1 M in 2 hours and from 0.2 M to 0.05 M in 4 hours. The order of the reaction is—

- (1) Zero (2) Two
 (3) One (4) Half

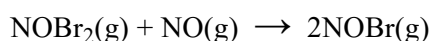
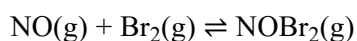
82. Which of the following is not a characteristics of lyophilic colloids :

- (1) Viscosity is higher than dispersion medium
- (2) Colloid particles are heavily hydrated
- (3) Irreversible in nature
- (4) They are easily formed by direct mixing

83. The rate constant for the first order reaction is given by : $\log K = 14.34 - \frac{1.25 \times 10^4}{T}$ Calculate the activation energy :

- (1) 2.393×10^5 J
- (2) 8.69×10^4 J
- (3) 1.18×10^5 J
- (4) 8.69×10^5 J

84. The following mechanism has been proposed for the reaction of NO with Br_2 to form NOBr :



If the second step is the rate determining step, the order of the reaction with respect to NO(g) is

- (1) 2
- (2) 1
- (3) 0
- (4) 3

85. **Assertion (A) :** The formation of micelles takes place only above a particular concentration called CMC

Reason (R) : On dilution micelles revert back to ions.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

SECTION-B (CHEMISTRY)

86. Which of the following complex is anion :-

- (1) Penta carbonyl iron (0)
- (2) Fluoro pentaamine cobalt (III)
- (3) Trioxalato ferrate (III)
- (4) Dichloro diammine platinum (II)

87. Which of the following is not correct ?

- (1) CFSE in weak field ligand for d^4 configuration is $-0.6 \Delta_0$
- (2) Effective atomic number of Pt in $[\text{PtCl}_6]^{-2}$ is 84
- (3) Br^- is a weak field ligand than H_2O
- (4) $[\text{Cr}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ shows linkage isomerism

88.

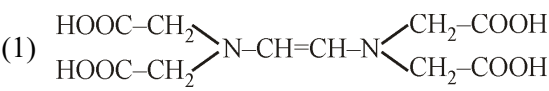
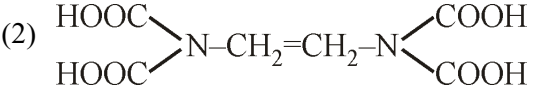
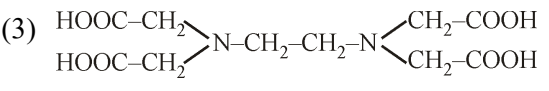
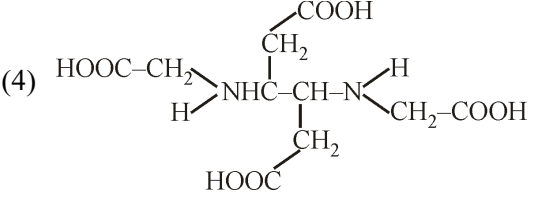
Column I		Column II	
(A)	CO	(P)	Monodentate ligand
(B)	En	(Q)	Anionic ligand
(C)	Gly	(R)	Form organometallic compound with metal
(D)	Thio cyanato	(S)	Bidentate ligand
		(T)	Neutral ligand

Select the correct option-

- (1) $A \rightarrow P, R, T; B \rightarrow S, T; C \rightarrow Q, S; D \rightarrow P, Q$
- (2) $A \rightarrow P, R; B \rightarrow S, T; C \rightarrow Q, R, S; D \rightarrow P, Q, R$
- (3) $A \rightarrow P, R, T; B \rightarrow S, T; C \rightarrow Q, S; D \rightarrow P, Q, R, S$
- (4) $A \rightarrow R, T; B \rightarrow S, T; C \rightarrow Q, R, S; D \rightarrow P, Q$

89. Which of the following is not correctly matched ?

- (1) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ - Coloured, square planar
- (2) $[\text{Fe}(\text{ox})_3]^{3-}$ - Paramagnetic, show two optical isomers
- (3) $[\text{AuCl}_4]^-$ - square planar, Diamagnetic
- (4) $[\text{Ni}(\text{dmg})_2]$ - Tetrahedral, Inter H-bonding present

90. The correct structure of ethylenediaminetetraacetic acid (EDTA) is:
- (1) 
- (2) 
- (3) 
- (4) 
91. Consider the following complex ions, P, Q and R. $P = [FeF_6]^{3-}$, $Q = [V(H_2O)_6]^{2+}$ and $R = [Fe(H_2O)_6]^{2+}$. The correct order of the complex ions, according to their spin-only magnetic moment values (in B.M.) is :
- (1) $R < Q < P$ (2) $Q < R < P$
 (3) $R < P < Q$ (4) $Q < P < R$
92. In a body centred cubic unit cell of a metal edge length is 4×10^{-10} m. The diameter of the atom will be :-
- (1) 3.46×10^{-10} m
 (2) 2×10^{-10} m
 (3) 1×10^{-10} m
 (4) 1.732×10^{-10} m
93. The total number of octahedral void(s) per atom present in a cubic close packed structure is :-
- (1) 1 (2) 2 (3) 3 (4) 4
94. For antifluorite type solid $r_+ + r_-$ is equal to:
- (1) a (2) $\frac{a}{2}$
 (3) $\frac{a\sqrt{3}}{4}$ (4) $\frac{a\sqrt{3}}{2}$
95. The number of unit cells in 58.5 g of NaCl is nearly:-
- (1) 6×10^{20} (2) 3×10^{22}
 (3) 1.5×10^{23} (4) 0.5×10^{24}
96. Which of the following is incorrect for lead storage battery:
- (1) Electrolyte is 38% H_2SO_4 solution.
 (2) During discharging conc. of H_2SO_4 decreases.
 (3) This is a secondary cell.
 (4) During charging $PbSO_4$ is oxidised into PbO_2 at cathode.
97. In the following reaction $A \rightarrow B + C$, rate constant is 0.001 Ms^{-1} . If we start with 1 M of A then concentration of A and B after 10 minutes are respectively
- (1) 0.5 M, 0.5 M (2) 0.6 M, 0.4 M
 (3) 0.4 M, 0.6 M (4) None of these
98. In a certain gaseous reaction between X and Y, $X + 3Y \rightarrow XY_3$. The initial rates are reported as follows:
- | [X] | [Y] | Rate |
|-------|-------|-------------------------|
| 0.1 M | 0.1 M | 0.002 Ms^{-1} |
| 0.2 M | 0.1 M | 0.002 Ms^{-1} |
| 0.3 M | 0.2 M | 0.008 Ms^{-1} |
- The rate law is :
- (1) $r = K[X][Y]^3$ (2) $r = K[X]^0[Y]^2$
 (3) $r = K[X][Y]$ (4) $r = [X]^0[Y]^3$
99. For a zero order reaction $t_{75\%} = 30$ min. Find $t_{50\%}$:-
- (1) 15 min (2) 20 min
 (3) 30 min (4) 75 min
100. For the reaction $2A \rightarrow B + 3C$;
 if $-\frac{d[A]}{dt} = K_1[A]^2$; $\frac{d[B]}{dt} = k_2[A]^2$; $\frac{d[C]}{dt} = k_3[A]^2$ the correct relation between k_1 , k_2 and k_3 is :-
- (1) $k_1 = k_2 = k_3$ (2) $2k_1 = k_2 = 3k_3$
 (3) $4k_1 = k_2 = 3k_3$ (4) $\frac{k_1}{2} = k_2 = \frac{k_3}{3}$

Topic : MOLECULAR BASIS OF INHERITANCE, ECOLOGY FROM PREDATIONS [BACK UNIT] : HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH

SECTION-A (BIOLOGY-I)

101. According to Gause's competitive exclusion principal, two species cannot simultaneously occupy the same ecological :-

- (1) Habitat (2) Community
(3) Niche (4) Ecosystem

102. Select the incorrect statement in the following :-

- (1) Biological control methods adopted in agricultural pest control are based on the ability of the predator to regulate prey population
(2) When Connell experimentally removed the *Balanus* he observed that surface area occupied by the *Chathamalus* in intertidal zone decreased.
(3) In the rocky intertidal communities of the American pacific coast the starfish *Pisaster* is an important Predator.
(4) Nearly 25% of all insects are known to be phytophagous.

103. Breeding season of cuckoo is :-

- (1) Spring to summer (2) Summer to spring
(3) Autumn to winter (4) Winter to summer

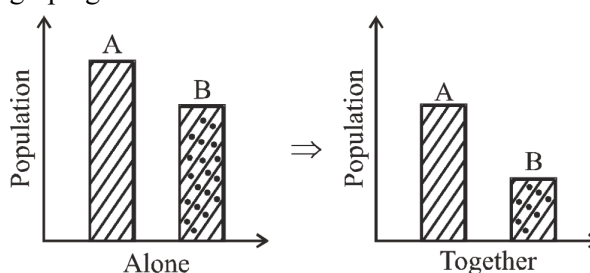
104. Which is/are correct ?

- (1) Interaction of biotic and abiotic components result in a physical structure that is characteristic for each type of community.
(2) Interaction of biotic and abiotic components result in a physical structure that is characteristics for each ecosystem.
(3) Interaction of biotic and abiotic component result in a physical structure that is characteristics of ecology.
(4) Interaction of biotic and abiotic component result in a physical structure that is characteristics of population.

105. Which of the following example is an exception to the ecological rule of competitive exclusion ?

- (1) Abingdon tortoise on galapagos island.
(2) *Balanus* and *chathamalus*.
(3) Warbler's bird species on pine tree.
(4) All of the above

106. Choose the correct option for the relation between species A & species B shown in the graph given below :-



- (1) Commensalism
(2) Mutualism
(3) Competition
(4) Amensalism

107. Which one is correct :-

	I		II
(1)	Formation of dark coloured amorphous substance	(i)	Fragmentation
(2)	Water soluble nutrients go down into the soil horizon	(ii)	Humification
(3)	release of inorganic nutrient by microbial activities	(iii)	Mineralisation
(4)	Break down detritus into smaller particles	(iv)	Leaching

- 108.** Ecosystem is a dynamic system because continuous interaction is going on in between biotic or abiotic components so ecosystem is present in an equilibrium position. Ecosystem is also self maintainable and self regulatory system, it means an ecosystem maintain a balance in between different trophic levels. It is called :-
- (1) Biosphere (2) Homeostasis
(3) Ecology (4) Energy flow
- 109.** How many statements are correct ?
- (a) Net primary productivity is available biomass for consumption to heterotrops
(b) Secondary productivity is defined as rate of formation of new organic matter by heterotrophs
(c) The annual net primary productivity of ocean is greater than land
(d) Warm and moist environment favours decomposition
- (1) a,b,c,d (2) a,b,c
(3) a,b,d (4) a,c
- 110.** Some species have evolved the ability to regulate, but only over a limited range of environmental conditions beyond which they are simply
- (1) Regulator
(2) Conformers
(3) Migrater
(4) None of these
- 111.** Earth is an :
- (1) Open system for energy and open system for minerals
(2) Closed system for minerals and open system for energy
(3) Closed system for energy and open system for minerals
(4) Closed system for energy and closed system for minerals
- 112.** How many characteristics in the list given below are structural :-
Dominance, Energy use efficiency, Species diversity, Species composition, stratification, Organism size, Nutrient conservation, Food web complexity, physiognomy.
- (1) Three (2) Four (3) Five (4) Six
- 113.** How much of carbon is fixed in the biosphere through photosynthesis annually ?
- (1) 4×10^{13} kg (2) 3×10^{13} kg
(3) 4×10^{17} kg (4) 5×10^{47} kg
- 114.** Ecosystems present on earth need a constant supply of energy :-
- (1) To synthesis the molecules they require
(2) To counter act the universal tendency toward increasing disorderlines.
(3) For the continuity of life on earth
(4) All of these
- 115.** Some nutrients gets attached to the biomass of microbes which becomes unavailable to other organisms, it is known as :-
- (1) Acclimatization
(2) Leaching
(3) Mineralization
(4) Nutrient immobilization
- 116.** "To meet with the deficit which occurs due to imbalance in the rate of influx and efflux" is the function of :-
- (1) Biosphere (2) Reservoir
(3) Solar energy (4) Species diversity
- 117.** The amount of Nutrients, such as carbon, Nitrogen, phosphorus, calcium, present in the soil at any given time is referred as :-
- (1) Standing crop
(2) Standing state
(3) Biogeochemical cycling
(4) Trophic level

118. Read the following and find out which one is incorrect about ecological pyramids :-

S.No.	Name	Number Pyramid	Biomass Pyramid	Energy Pyramid
1	Grassland			
2	Tree			
3	Ocean			
4	Lake			

119. Bamboo plant is growing in a far forest than what will be the trophic level of it :-

- (1) First trophic level
- (2) Second trophic level
- (3) Third trophic level
- (4) Fourth trophic level

120. Small animals have a larger surface area relative to their volume, thus when there is cold outside :-

- (1) They lose body heat very slowly and easily regulate internal body temperature
- (2) They lose body heat very fast and easily regulate internal temperature
- (3) They lose body heat very fast and find it difficult to regulate internal body temperature
- (4) They lose body heat very slowly and easily conform to environmental condition

121. A physiological adaptation among the following is :-

- (1) Low respiratory rate at high altitude
- (2) Higher RBC count in people living in the plains
- (3) Increase in binding capacity of Hb with oxygen
- (4) Increase in red blood cell production at high altitude

122. Read the following statement and find out how many statements are correct and choose correct answer :-

- (a) A few organisms can tolerate and thrive in a wide range of temperature, they are called eurythermal
- (b) Vast majority of organisms are restricted to a narrow range of temperature, such organism are called stenothermal.
- (c) The productivity and distribution of plants is also heavily dependent on water
- (d) The salt concentration is less than 5 ppt in inland waters, 30-35 ppt in the sea and > 100 ppt in some hypersaline lagoons.

- (1) a,d only
- (2) a,b,c only
- (3) d only
- (4) a,b,c,d

123. Which one is correct :-

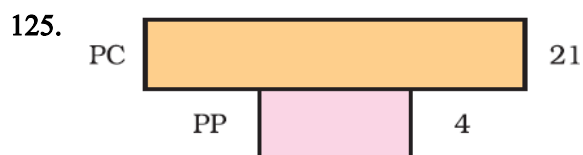
(1)	Cold area animals	Small in size	Large surface area	Low volume
(2)	Warm area animals	Small in size	Low surface area	Large volume
(3)	Warm area animals	Small in size	Large surface area	Low volume
(4)	Cold area animals	Large in size	Small surface area	Small volume

124. If you had ever been to any high altitude place like Rohtang Pass near Manali and Mansarovar. What kind of changes you are supposed to have while staying at same place for 1 year :-

- (a) Increased breathing rate
- (b) Nausea, Fatigue, Palpitations
- (c) Increased R.B.C. number
- (d) Decreased R.B.C. size

How many above statements are correct ?

- (1) statements a,b,c,d
- (2) statements a,c
- (3) statements a,d
- (4) statements b,c



Above given pyramid can represent :-

- (1) Pyramid of biomass is grassland.
 - (2) Number of insects feeding on big tree.
 - (3) Pyramid of biomass in lake.
 - (4) Pyramid of energy in forests.
126. Which of the following statement is false ?
- (1) Spermiogenesis is the process of formation of sperms from spermatogonia.
 - (2) During pregnancy the level of thyroxine is increased in the maternal blood.
 - (3) 200 million secondary spermatocytes are required to form 400 million spermatozoa.
 - (4) Oogenesis is initiated during embryonic stage of a female.
127. How many X-chromosomes are present in the 2nd polar body in human beings ?
- (1) 0 (2) 23 (3) 2 (4) 1
128. Which of the following statement is true ?
- (1) Acrosome of sperm serves no particular function.
 - (2) Menstrual fluid can easily clot.
 - (3) The part of the fallopian tube closest to the ovary is ampulla.
 - (4) Viability of sperm is determined by it's motility.
129. In the given list secretions of how many structures are essential for maturation and motility of sperms ? Epididymis, vas deferens, seminal vesicle, prostate
- (1) 4 (2) 3 (3) 2 (4) 1
130. Which pair is not matched correctly ?
- (1) Oviduct $\Rightarrow$ 10-12 cm long
 - (2) Testes $\Rightarrow$ 4-5 cm in length
 - (3) Ovary $\Rightarrow$ 2-4 cm in length
 - (4) Sperms per ml. of normal semen $\Rightarrow$ 200-300 million
131. The functions of male accessory ducts and glands are maintained by ?
- (1) Testicular hormones $\rightarrow$ LH and FSH
 - (2) Inhibin
 - (3) Pituitary hormone $\rightarrow$ Androgens
 - (4) Testicular hormones $\rightarrow$ Androgens
132. Lack of menstruation may be indicative of ?
- (1) Pregnancy (2) Stress
 - (3) Poor health (4) All of the above
133. Which of the following statement is incorrect ?
- (1) During pregnancy all events of menstrual cycle stops and there is no menstruation.
 - (2) Menstruation only occurs if the released ovum is not fertilised.
 - (3) In presence of fertilisation, the corpus luteum starts degenerating.
 - (4) During ovulation, only one ovum is released per menstrual cycle
134. Chorion is formed of ?
- (1) Mesoderm on outer side and ectoderm on inner side.
 - (2) Mesoderm and endoderm.
 - (3) Ectoderm on outer side and mesoderm on inner side.
 - (4) Ectoderm and endoderm.
135. Which of the following is not included in invivo fertilisation ?
- (1) GIFT (2) IUI (3) AI (4) IUT

SECTION-B (BIOLOGY-I)

136. Solar energy converted to chemical energy by plant and chemical energy get converted in to kinetic or heat energy in animals, it shows :-
- (1) 1st law of thermodynamic in food chain
 - (2) 2nd law of thermodynamic in food chain
 - (3) Above both laws in food chain
 - (4) Above statement is not related with law of thermodynamic

137. Several processes occur simultaneously during decomposition of detritus. Choose incorrect option for catabolism and fragmentation :-

	Features	Catabolism	Fragmentation
A	Organism involved	Saprotrophs	Detritivores
B	Example	Bacteria	Earthworm
C	Function	Breakdown of large organic matter	Organic substance change in to inorganic

- (1) A and B (2) C only
(3) B and C (4) A only
138. How many organism in the list given below are microconsumers?
Phytoplankton, Grass, Tree, Zooplankton, Grasshopper, Cow, Bird, Fish, Wolf, Diatoms, Bacteria, Fungi.

(1) Two (2) Ten (3) Six (4) Five

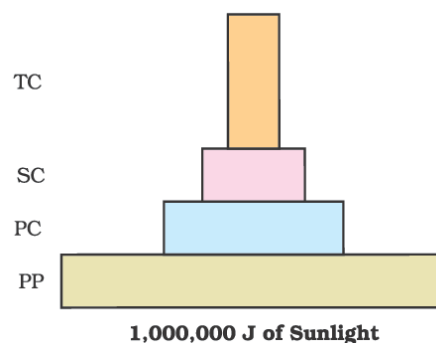
139. Read the following statements and find out the incorrect statements :-

- (1) Carbon constitutes 49 percent of dry weight of organisms and is next to water.
(2) Of the total quantity of global carbon, 71 percent carbon is found dissolved in ocean.
(3) Atmosphere contains about 10 percent of total global carbon.
(4) The oceanic reservoir regulates the amount of carbon dioxide in the atmosphere.

140. The phosphorus cycle differs from the carbon cycle in that :-

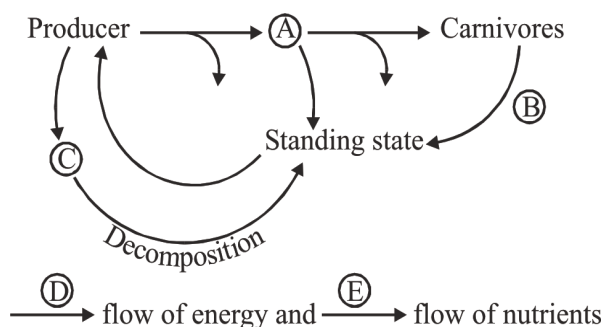
- (1) the phosphorus does not enter living organisms, whereas carbon does
(2) the phosphorus cycle does not include a gaseous phase, whereas the carbon cycle does
(3) the phosphorus cycle includes a solid phase, whereas the carbon cycle does not
(4) the primary reservoir of the phosphorus cycle is the atmosphere, whereas the primary reservoir for the carbon cycle is in rock.

141. In the given pyramid calculate the amount of energy present at secondary consumer level.



- (1) 100
(2) 1000
(3) 5000
(4) 10

142. Match the following :-

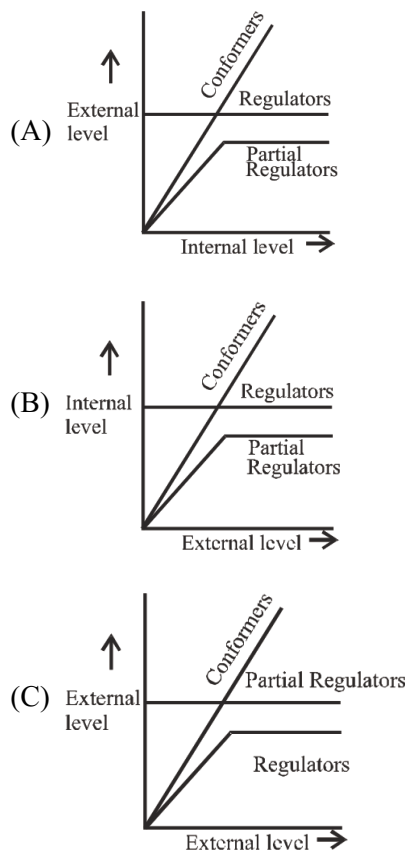


	A	B	C	D	E
(1)	Herbivores	Detritus	Litter	Cyclic	Linear
(2)	Key industry animals	Decomposition	Litter	Linear	Cyclic
(3)	T-2	Duff	GFC	Cyclic	Linear
(4)	Bacteria	Predation	Fungi	Linear	Linear

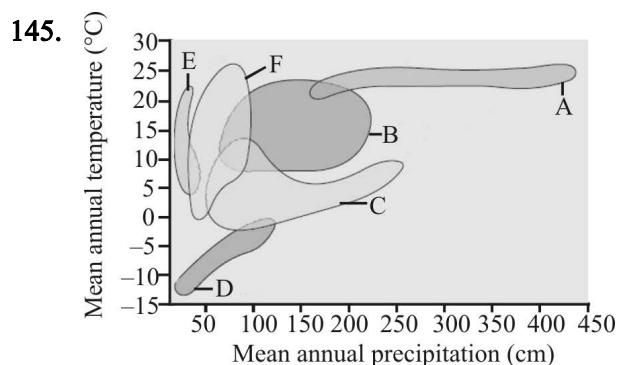
143. Just as a person moving from delhi to shimla to escape the heat for the duration of hot summer, thousands of migratory birds from siberia and other extremely cold nothern regions move to :-

- (1) Western ghats
(2) Meghalaya
(3) Corbett national park
(4) Keoladeo national park

144. The figure below shows three diagrammatic representations (A,B,C) of organismic response, Identify the correct diagram :-



- (1) A, B, C (2) A, B
(3) Only B (4) A, C



Identify the biome A B C D from above graph and choose the correct option in respect to their vegetation :-

	A	B	C	D
(1)	Alpine	Acacia	Oak	Prosopis
(2)	Oak	Sal, Teak	Pine	Mosses
(3)	Sal, Teak	Pine	Oak	Lichen
(4)	Hopea	Oak	Pine	Alpine

146. Which of the following is not included in the natural method of contraception ?

- (1) Saheli
(2) Lactational amenorrhoea
(3) Withdrawal method
(4) Rhythm method

147. In which of the following technique zygote or early embryo (with upto 8 blastomeres) is transferred into the fallopian tube.

- (1) GIFT
(2) ZIFT
(3) AI
(4) IUT

148. Which of the following pair is matched correctly ?

- (1) Hormone releasing IUD $\Rightarrow$ Multiload 375
(2) Copper releasing IUD $\Rightarrow$ LNG 20
(3) Non medicated IUD $\Rightarrow$ Progestasert
(4) Combination pills $\Rightarrow$ Mala-D

149. Which of the following statement is correct ?

- (1) 'Nirodh' is a popular brand of condom for female.
(2) During Lactational amenorrhoea, level of prolactin is very low.
(3) In a menstrual cycle of 28 days, fertile period is from 10 to 17 days.
(4) Chorionic villus sampling can not be done before 16th week of pregnancy

150. How many in the list given below are secreted by sertoli cells :

ABP, LH, FSH, AMH, Inhibin

- (1) One
(2) Three
(3) Two
(4) Four

Topic : MOLECULAR BASIS OF INHERITANCE, ECOLOGY FROM PREDATIONS [BACK UNIT] : HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH

SECTION-A (BIOLOGY-II)

151. Emergency contraceptives include which of the following ?

- (1) Administration of Progestogens
- (2) Administration of Progestogens estrogen combination
- (3) IUD
- (4) All of the above

152. **Assertion (A)** : Secondary spermatocyte has only 23 chromosomes.

Reason (R) : Spermatogonia multiply by mitotic division.
Choose the correct answer from the option given below :

- (1) Both (A) & (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A)

153. Sugar fructose is present in the secretion of :

- (1) Prostate gland
- (2) Bulbourethral gland
- (3) Seminal vesicle
- (4) Bartholin's gland

154. Identify the correct match from Column I, II and III.

Column I		Column II		Column III	
(A)	Ovulatory phase	(a)	6 th - 13 th day	(i)	Breakdown of endometrium
(B)	Menstruation phase	(b)	15 th - 28 th day	(ii)	Release of secondary oocyte
(C)	Secretory phase	(c)	14 th day	(iii)	Development of mature follicle
(D)	Follicular phase	(d)	1 st -5 th day	(iv)	Formation of corpus luteum

- (1) A-c-i, B-d-ii, C-a-iii, D-b-iv
- (2) A-c-ii, B-a-i, C-b-iii, D-d-iv
- (3) A-d-i, B-a-iv, C-d-iv, D-c-ii
- (4) A-c-ii, B-d-i, C-b-iv, D-a-iii

155. **Assertion (A)** : Degeneration of endometrium leads to menstruation.

Reason (R) : In the absence of fertilisation, the corpus luteum is maintained by HCG hormone.

- (1) Both (A) & (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A)

156. **Statement-I** : The embryo with 8 to 16 blastomeres is called a morula.

Statement-II : The blastomeres in the blastocyst are arranged into an outer layer called trophoblast.

Choose the correct answer from the option given below.

- (1) Both statement-I and statement-II are correct
- (2) Statement-I is correct but statement-II is incorrect
- (3) Statement-I is incorrect but statement-II is correct
- (4) Both statement-I and statement-II are incorrect

157. What is the effect of menopause on the levels of FSH, LH, estrogen and GnRH :-

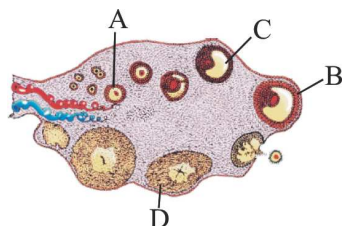
- (1) [FSH]↑; [LH]↑; [Estrogen]↓; [GnRH]↑
- (2) [FSH]↓; [LH]↓; [Estrogen]↓; [GnRH]↓
- (3) [FSH]↓; [LH]↓; [Estrogen]↑; [GnRH]↑
- (4) [FSH]↓; [LH]↑; [Estrogen]↓; [GnRH]↓

158. **Statement-I** : Parturition is induced by a complex neuroendocrine mechanism.

Statement-II : Placenta is jointly formed by chorionic villi and uterine tissue.

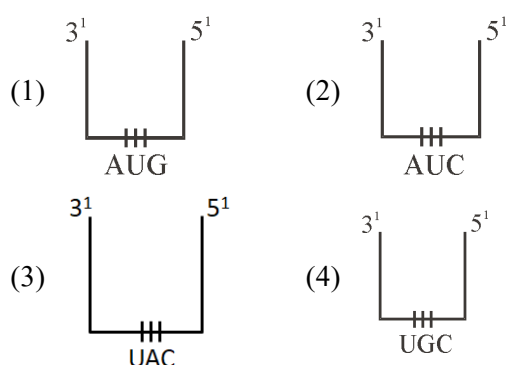
- (1) Both statement-I and statement-II are correct
- (2) Statement-I is correct but statement-II is incorrect
- (3) Statement-I is incorrect but statement-II is correct
- (4) Both statement-I and statement-II are incorrect

159. The figure given below shows section view of ovary which describe development of follicle (A, B, C and D). Select the option with correct identification.



	A	B	C	D
(1)	Secondary follicle	Tertiary follicle	Graafian follicle	Corpus luteum
(2)	Primary follicle	Graafian follicle	Tertiary follicle	Corpus luteum
(3)	Secondary follicle	Graafian follicle	Tertiary follicle	Corpus luteum
(4)	Primary follicle	Tertiary follicle	Graafian follicle	Corpus luteu

160. In a female undergoing hysterectomy, which one of the following event will not occur?
- (1) Formation of Graafian follicle
 - (2) Ovulation
 - (3) Menstruation
 - (4) Completion of meiosis-I and formation of secondary oocyte.
161. Which property of genetic code is explained by wobble hypothesis?
- (1) Degeneracy
 - (2) Non-overlapping
 - (3) Non-ambiguous
 - (4) Universal
162. Which of the following tRNA doesn't exist?



163. Which of the following steps in transcription is catalysed by RNA polymerase itself ?
- (1) Initiation
 - (2) Elongation
 - (3) Termination
 - (4) All of these
164. Sequence of N-bases in mRNA is (CUGA)₃, then how many types of amino acids will be present in the polypeptide chain which will be formed from this mRNA?
- (1) Four
 - (2) Three
 - (3) Two
 - (4) Five
165. If 8 types of nitrogenous bases are available in genetic code and 60 types of amino acids are required for protein synthesis, then genetic code is present in which form ?
- (1) Singlet
 - (2) Doublet
 - (3) Triplet
 - (4) Quadraplate
166. Polypeptide synthesis require energy during :
- P – An amino acid activation
Q – Binding of large ribosomal subunit
R – Movement of peptidyl tRNA to P site
S – Binding of amino acyl tRNA to A site
- (1) P, Q, R, S
 - (2) P, Q, R
 - (3) P, R, S
 - (4) Only P, S
167. If both the strands of DNA are transcribed then the two RNA molecules formed must be-
- (1) Antiparallel and identical
 - (2) Antiparallel and complementary
 - (3) parallel and identical
 - (4) parallel and complementary
168. Failure of segregation of chromatids results in the gain or loss of a chromosome(s) called :-
- (1) Euploidy
 - (2) Aneuploidy
 - (3) Polyploidy
 - (4) Diploidy

169. During translocation in translation process, tRNA attached with growing polypeptide chain shifts from:
- (1) P to E site (2) P to A site
(3) E to P site (4) A to P site
170. The accessibility of promoter regions of prokaryotic DNA by RNA polymerase is in many cases regulated by the interaction of some protein with sequence termed as :
- (1) promoter
(2) operator
(3) regulator
(4) cistron
171. Given below are the various steps of DNA-fingerprinting, arrange these steps in correct sequence :-
- (i) Isolation of DNA
(ii) Hybridisation using Labelled VNTR probe
(iii) Transferring (blotting) of separated DNA fragments to synthetic membrane, such as nitrocellulose or nylon.
(iv) Digestion of DNA by restriction endonuclease
(v) Detection of hybridised DNA fragment by autoradiography.
(vi) Separation of DNA fragments by electrophoresis.
- (1) i, iv, iii, vi, ii, v (2) i, ii, iii, iv, v, vi
(3) i, iv, vi, iii, ii, v (4) i, ii, iv, v, vi, iii
172. *E.coli* cells with a mutated Z gene of the lac operon cannot grow in medium containing only lactose as the source of energy because :-
- (1) The lac operon is constitutively active in these cells
(2) They cannot synthesize functional β -galactosidase
(3) In the presence of glucose *E.coli* cells do not utilize lactose
(4) They cannot transport lactose from the medium into the cell
173. In eukaryotes the regulation of gene expression could be exerted at :-
- I. Transcriptional level
II. Processing level
III. Transport of ribosomal subunits from nucleus to the cytoplasm
IV Translational level
- Choose the correct combination for given statements:-
- (1) Only I, II, IV
(2) Only IV
(3) I, II, III, IV
(4) Only II, III
174. Match column-I with column-II and select the correct option from the codes given below.
- | Column-I
(Codons) | | Column-II (Translated amino acid) | |
|----------------------|-----|-----------------------------------|---------------|
| (A) | UUU | (i) | Serine |
| (B) | GGG | (ii) | Methionine |
| (C) | UCU | (iii) | Phenylalanine |
| (D) | CCC | (iv) | Glycine |
| (E) | AUG | (v) | Proline |
- (1) A-(iii), B-(iv), C-(i), D-(v), E-(ii)
(2) A-(iii), B-(i), C-(iv), D-(v), E-(ii)
(3) A-(iii), B-(iv), C-(v), D-(i), E-(ii)
(4) A-(ii), B-(iv), C-(i), D-(v), E-(iii)
175. In which one of the following combination (1-4) of number of chromosome in octaploid tritcale (56) correctly represented :-
- | Combination | Monosomic | Haploid | Nullisomic | Trisomic |
|-------------|-----------|---------|------------|----------|
| (1) | 55 | 28 | 54 | 57 |
| (2) | 7 | 28 | 29 | 27 |
| (3) | 55 | 21 | 54 | 55 |
| (4) | 21 | 41 | 40 | 4 |

176. If the sequence of coding strand of transcription unit is 5' ATG CAT GCA TGC ATG CAT 3' what will be the sequence of m-RNA ?

- (1) 5' UAG GUA CGU ACG UAC GUA 3'
- (2) 5' AUG CAU GCA UGC AUG CAU 3'
- (3) 3' AUG CAU GCA UGC AUG CAU 5'
- (4) 3' UAC GUA CGU ACG UAC GUA 3'

177. Find out the correct match :-

(i)	RNA polymerase-I	(a)	hnRNA
(ii)	RNA polymerase-II	(b)	SnRNA
(iii)	RNA polymerase-III	(c)	18s rRNA

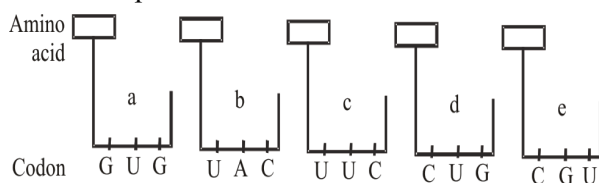
- (1) i-b, ii-c, iii-a (2) i-c, ii-a, iii-b
- (3) i-c, ii-b, iii-a (4) i-b, ii-a, iii-c

178. Consider the following four statements (a-d) and select the option which include all the correct ones only:-

- (a) Transcription and translation can be coupled in bacteria
- (b) The split gene arrangement represents the advanced feature of the genome.
- (c) The process of splicing represents the dominance of RNA-world
- (d) The presence of introns in split gene is reminiscent of antiquity

- (1) a, b and d (2) a, b and c
- (3) a, c and d (4) b, c and d

179. Find the correct sequence of binding of the following amino-acyl-tRNA complex during translation of a mRNA transcribed by a DNA segment having following base sequence 3' TAC GTG CTG TTC CGT 5' Template strand. Select the correct answer showing the correct order of alphabet



- (1) b, a, d, c, e (2) a, b, c, d, e
- (3) d, c, b, a, e (4) d, b, c, a, e

180. **Assertion (A)** : Mutation results in changes in genotype and phenotype of an organism.

Reason (R) : Mutation, leads to alteration of DNA sequences that is variations in DNA.

Choose the correct answer from the options given below.

- (1) Both (A) and (R) are correct and (R) is correct explanation of (A)
- (2) Both (A) and (R) are correct and (R) is not correct explanation of (A)
- (3) (A) is correct (R) is not correct.
- (4) (A) is not correct (R) is correct.

181. **Assertion** : In the first phase of translation itself amino acids are activated in the presence of ATP

Reason : During translation for the formation of a peptide bond energy is required.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

182. Read the following statements :-

- (i) Eukaryotic mRNA is polycistronic whereas prokaryotic mRNA is monocistronic
- (ii) Prokaryotic mRNA is capless and tail-less whereas eukaryotic mRNA is with cap and tail
- (iii) The precursor of prokaryotic mRNA is hnRNA
- (iv) In replication only a segment of DNA and only one of the strand is copied into RNA
- (v) In eukaryotes, transcription occurs in nucleus and translation in cytoplasm

Which of the above statements are correct ?

- (1) (ii), (iii) and (v) (2) (iii), (iv) and (v)
- (3) (ii) and (v) (4) (iii) and (iv)

183. Read the following statements-

- (i) Promoter and terminator flank the structural gene
- (ii) Promoter is located at 3' end of coding strand upstream of the structural gene
- (iii) In lac operon, y-gene codes for permease which increases permeability of cell membrane to β -galactosides.

How many statements are correct ?

- (1) One
- (2) Two
- (3) Three
- (4) None

184. In a polypeptide or a protein, amino acids are linked by a peptide bond which is formed when the :-

- (1) Carboxyl ($-COOH$) group of one amino acid reacts with the amino ($-NH_2$) group of the next amino acid with the addition of a water moiety
- (2) Carboxyl ($-COOH$) group of one amino acid reacts with the amino ($-NH_2$) group of the next amino acid with the elimination of a water moiety
- (3) Amino ($-NH_2$) group of one amino acid reacts with the carboxyl ($-COOH$) group of the next amino acid with the elimination of a water moiety
- (4) Amino ($-NH_2$) group of one amino acid reacts with the carboxyl ($-COOH$) group of the next amino acid with the addition of a water moiety

185. Read the following four statements (A-D) :-

- (A) Newly arrived t-RNA with amino acid binds at A-site except at initiation codon.
- (B) Reverse transcription is catalysed by RNA dependent DNA polymerase.
- (C) Anti-shine delgarno sequence is present in larger subunit of ribosome.
- (D) The enzyme that catalyses the peptide bond formation is located in smaller ribosomal subunit.

How many of the above statement are correct ?

- (1) Four
- (2) Three
- (3) Two
- (4) One

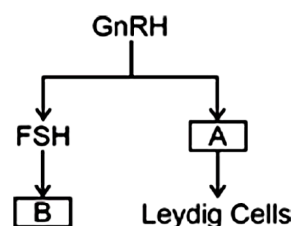
SECTION-B (BIOLOGY-II)

186. The function of ovary is:

- a. To produce female gamete
- b. To provide the site for fertilisation.
- c. To provide the site for implantation.
- d. To produce steroid hormones.

- (1) a and b
- (2) a, b and d
- (3) a, b and c
- (4) a and d

187. Study the given flowchart.



Select the option which correctly answers A and B.

	A	B
(1)	LH	Sertoli cells
(2)	FSH	Follicular cells
(3)	FSH	Sertoli cells
(4)	Testosterone	Testis

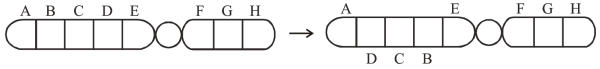
188. At the time of puberty how many primary follicles are left in a female ?

- (1) 60,000 – 80,000
- (2) 40,000 – 60,000
- (3) 1,20,000 – 1,60,000
- (4) 80,000 – 1,20,000

189. Milk is secreted by the alveoli present in the mammary lobes and it reach upto nipple through following regions. Arrange these regions in a correct order.

- a. Lactiferous duct
- b. Mammary duct
- c. Ampulla
- d. Mammary tubule

- (1) $b \rightarrow c \rightarrow d \rightarrow a$
- (2) $d \rightarrow b \rightarrow c \rightarrow a$
- (3) $b \rightarrow d \rightarrow c \rightarrow a$
- (4) $a \rightarrow b \rightarrow d \rightarrow c$

- 190.** Which of the following gland secretion is not the part of seminal plasma?
- (1) Seminal vesicle (2) Prostate
(3) bulbourethral (4) Bartholin's
- 191.** Which r-RNA is responsible for the attachment of ribosome with t-RNA ?
- (1) 16s r-RNA (2) 18s r-RNA
(3) 5.8s r-RNA (4) 5s r-RNA
- 192.** rRNA that catalyse the formation of peptide bond in eukaryotes is synthesised by
- (1) RNA Polymerase I
(2) RNA Polymerase II
(3) RNA Polymerase-III
(4) RNA -Polymerase
- 193.** The UTR is present at which end of m-RNA :-
- (1) only at 5' end (2) only at 3' end
(3) Both 5' & 3' end (4) After starting codon
- 194.** Which statement is true for transcription?
- (1) Both strands of DNA are copied during transcription.
(2) Total DNA of an organism get transcribed
(3) Adenine now form base pair with cytosine instead of thymine
(4) In prokaryotes sigma factor help RNA polymerase to bind to promoter.
- 195.** During RNA processing polyadenylation at 3' end of RNA occurs :-
- (1) by RNA polymerase - template dependent manner
(2) by poly A polymerase - template dependent manner
(3) by poly A polymerase - template independent manner
(4) by polynucleotide phosphorylase - template independent manner
- 196.** The given sequence when translated must form a polypeptide of how many amino acids ?
5' – CUCAAUCCAUGCCCUGGGCUUGUUAGCGA – 3'
- (1) Eight (2) Nine
(3) Six (4) Five
- 197.** Regulation of lac operon by repressor is referred to as–
- (1) Positive regulation
(2) Negative regulation
(3) Both (1) and (2)
(4) None
- 198.** Basis of DNA fingerprinting is :-
- (1) high degree of polymorphism in non coding sequences
(2) low degree of polymorphism in non coding sequences
(3) intermediate degree of polymorphism in coding sequences
(4) sequences showing no polymorphism
- 199.** Which of the following is true for Lac-operon :
- (i) Inactive repressor is formed
(ii) Operates in anabolic pathway
(iii) Inducer, inactivates repressor protein
(iv) Three structural genes z,y,a are involved
(v) i-gene synthesises repressor constitutively
- (1) i, ii, iii, iv (2) ii, iii, iv
(3) ii, iii, v (4) iii, iv, v
- 200.** Given below is a representation of kind of chromosomal mutation. What is the kind of mutation represented ?
- 
- (1) Deletion
(2) Duplication
(3) Paracentric inversion
(4) Pericentric inversion

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MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
05 18-06-2023	SYLLABUS COVERED FROM 22-05-2023 TO 17-06-2023	P	ELECTROSTATICS & CURRENT ELECTRICITY
		C	PERIODIC TABLE
		B	LIFE CYCLE OF ANGIOSPERM

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